

Quiz 4 Class 2 Solutions

Computer Security Fundamentals Spring 2026

Instructions: Answer the questions, **closed-book, closed-notes**. Time: 10 minutes

1. (**T**) When using Address Space Layout Randomization (ASLR), the relative distance (offset) between two functions within the same executable remains constant. (**True/False, 2 pts**)

2. (**B**) If a system administrator wants to ensure that a specific background process cannot consume more than 1GB of RAM, which Linux feature should they use? (**2 pts**)

- A. Linux Namespaces.
- B. Control Groups (cgroups).
- C. Seccomp (Secure Computing).
- D. Chroot "Jail".

3. (**B**) What is the primary motivation for an attacker to use Return-Oriented Programming (ROP) instead of traditional shellcode injection? (**2 pts**)

- A. ROP is easier to write because it uses standard assembly instructions.
- B. ROP can execute even if the stack and heap are marked as non-executable (DEP).
- C. ROP gadgets automatically bypass Address Space Layout Randomization (ASLR).
- D. ROP does not require a vulnerability like a buffer overflow to succeed.

4. (**Short Answer, 4 pts**) An attacker prepares the following shellcode to exploit a buffer overflow vulnerability that uses the strcpy() function:

```
char shellcode[] =  
"\x48\x31\xd2\x48\xbb\x2f\x2f\x00\x00\x00\x62\x69\x6e\x2f\x73\x68\x48\xc7\xc0\x3b";
```

Identify the issue with this shellcode and explain why the attack will likely fail. Note: You don't need to actually analyze the assembly instructions in the shellcode.

Solution: The issue is the null bytes (0x00) in the shellcode. The strcpy() function treats the null byte as a string terminator. strcpy() will stop copying data when it encounters the first null byte, this splices the shellcode and makes it non-functional.